

Zhangqi (Gavin) Duan

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Amherst, MA

Summary PhD student in Computer Science at UMass Amherst specializing in LLM training, alignment, and evaluation, with a focus on education technology and human-centered AI systems. Experienced in LLM fine-tuning (SFT), RL-based optimization (GRPO, DPO), and multitask training for interpretable model behavior. Published in leading venues (ACL, EMNLP, AIED) with a strong record of translating research prototypes into robust ML systems for real-world NLP and reasoning tasks.

Education University of Massachusetts Amherst

- PhD in Computer Science (Advisor: [Prof. Andrew Lan](#)), GPA: 3.83/4.00 Sep 2024 – Present
- MS in Computer Science, GPA: 3.82/4.00 Sep 2022 – May 2024
- BS in Computer Science & BS in Mathematics, GPA: 3.85/4.00 Sep 2018 – May 2022

Work UMass Amherst Center for Data Science

Amherst, MA

Experience Machine Learning Engineer Intern

May 2023 – Aug 2023

- Analyzed 6.5M tweets using scalable data processing methods to detect key patterns and outliers
- Applied Python Numpy, Pandas modules to perform advanced data cleaning and feature engineering resulted in 23% decrease in model runtime
- Built a Topic Model to categorize diverse topics, uncovering unknown events within Non-English textual data
- Implemented a User Model pipeline to target primary contributors to ongoing conversations with unsupervised learning algorithms (Spherical K-means clustering, hierarchical clustering)

Siemens Ltd

Beijing, China

Software Engineer Intern

May 2019 - Aug 2019

- Conducted database creation for 10,000 clients using SQL Subqueries/Merge Statement
- Resolved technical issues reported by clients or detected through internal monitoring
- Performed code reviews and provided feedback to team members

Publications Knowledge-Aligned Student Error Simulator for Open-Ended Coding Tasks

Zhangqi Duan, Nigel Fernandez, Andrew Lan

ACL, 2026

Automated Knowledge Component Generation for Interpretable Knowledge Tracing in Coding Problems

Zhangqi Duan, Nigel Fernandez, Arun Narayanan, Mohammad Hassany, Rafaella Alencar, Peter Brusilovsky, Bitu Akram, Andrew Lan

ACL Findings, 2026

Using LLMs for Knowledge Component-level Correctness Labeling in Open-ended Coding Problems

Zhangqi Duan, Arnav Kankaria, Dhruv Kartik, Andrew Lan

AIED, Short Paper, 2026

Test Case-Informed Knowledge Tracing for Open-ended Coding Tasks

Zhangqi Duan, Nigel Fernandez, Alexander Hicks, Andrew Lan

LAK, 2025

Multilingualism, Transnationality, and K-pop in the Online #StopAsianHate Movement

Tessa Masis, **Zhangqi Duan**, Weiai Wayne Xu, Ethan Zuckerman, Jane Yeahin Pyo, Brendan O'Connor

WebSci, 2025

README: Bridging Medical Jargon and Lay Understanding for Patient Education through Data-Centric NLP

Zonghai Yao, Nandyala Siddharth Kantu, Guanghao Wei, Hieu Tran, **Zhangqi Duan**, Sunjae Kwon, Zhichao Yang, README annotation team, Hong Yu

EMNLP Findings, 2024

Agent Performing Autonomous Stock Trading under Good and Bad Situations

Yunfei Luo, **Zhangqi Duan**

AI4ABM (ICLR Workshop), 2023

Research Experience **Persona-Grounded Student Simulation for Tutoring Dialogue** Feb 2026 – May 2026

- Formulated a history-aware student simulation task that models tutoring dialogue from both student knowledge and behavioral perspectives
- Developed a two-component RL-based framework for student simulation in tutoring dialogue, improving faithfulness by 4.2% in Act Matching and 3.8% in Error Matching
- Devised a **counterfactual DPO preference-pair construction mechanism** for the student simulator, enabling better persona alignment
- Optimized the persona generator through a custom **GRPO** reward function, refining student persona to improve downstream simulation performance

LLM Training with RL for Code Generation

Oct 2025 – Jan 2026

- Developed **RL-based** methodology (**GRPO**) for student code simulation, improving controllability and diversity of generated codes
- Designed a custom **RL reward** function combining code similarity metrics and error level IoU to better align generation with student error distributions
- Evaluated SFT-only vs. SFT+RL training at scale, yielding a **13.1%** improvement in error coverage
- Built an end-to-end LLM training and evaluation pipeline in PyTorch, supporting reproducible experimentation across reward variants and datasets

Multitask LLM Training for Interpretable Prediction and Generation Aug 2024 – Jan 2025

- Created the first LLM-based pipeline for knowledge component generation, validated against expert-level quality
- Designed a **soft-token conversion mechanism** to support end-to-end gradient flow through multiple models
- Developed KC-Gen KT, a **multitask learning** framework jointly optimizing code generation and correctness prediction, integrating interpretable student representations
- Achieved a **5%** AUC improvement on prediction tasks and a **4%** CodeBLEU gain on code generation tasks over state-of-the-art models, demonstrating the effectiveness of KC-Gen KT across diverse tasks

Retrieval-Augmented Lay Definition Generation in BioNLP

Sep 2021 – Jan 2022

- Improved patient comprehension of clinical notes using natural language generation
- Applied pretrained sequence-to-sequence models (BART) for controlled definition generation
- Built data pipelines to handle linguistic ambiguity and domain shift in medical text
- Deployed a full-stack NLP application using React and Django to deliver real-time explanations to end users

Teaching Experience	Teaching assistant, College of Info. and Computer Science CS 240: Reasoning Under Uncertainty • Assigned different tasks such as checking assignment and quiz correctness to other TAs and UCAs as head TA • Assisted in preparation of coursework and taught discussion sections, improving the learning experience for 200+ undergraduate students	Fall 2022, Spring 2023
Service	Mentoring Two undergraduate students, resulting in a publication at AIED 2026 Program Chair Member at AIED 2026	Sep 2025 – Present
Honors & Awards	Bay State Fellowship (Full Scholarship Sponsored by UMass CICS) Graduated Magna Cum Laude In MS from University of Massachusetts Amherst Graduated Magna Cum Laude In BS from University of Massachusetts Amherst Dean's List in all semesters	2022
Skills	Machine Learning: deep neural network architecture design, reinforcement learning algorithm implementation and reward/loss function design, preference alignment Large Language Models: prompt engineering and RAG, LLM-driven annotation and evaluation, fine-tuning and LoRA using SFT, DPO, GRPO Software Engineering: full-stack web development Programming Languages: Python, Java, JavaScript, HTML, CSS, SQL Frameworks: PyTorch, HuggingFace, TRL, vLLM, React, Node.js	